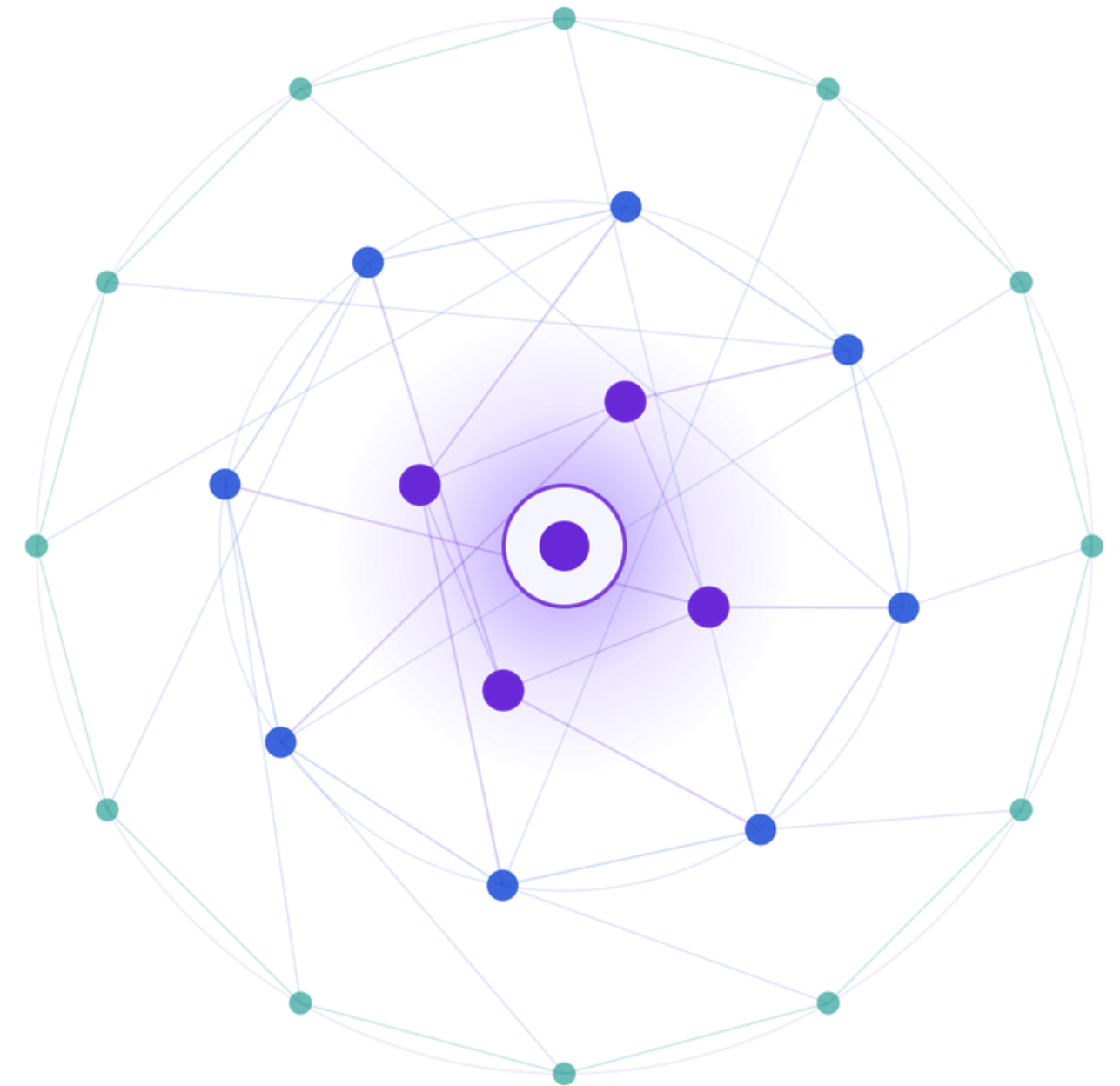


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 05
- Team Name - Pivot AI
- College Name - Thapar Institue of Engineering and Technology
- Member Name & Email 1- Jaivesh Chopra (jchopra_be24@thapar.edu)
- Member Name & Email 2- Zaid Yusuf (zyusuf_be24@thapar.edu)
- Member Name & Email 3- Ahaan Luther(aluther_be25@thapar.edu)
- Member Name & Email 4- Aakriti (abansal1_be24@thapar.edu)
- Submission Github link
<https://github.com/itsZaid05/CONTINUUM/tree/main>



Theme - 05

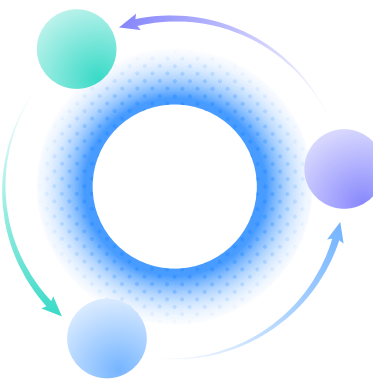
Interruptible - Real-Time Agents

Real-time agents that handle mid-task interruptions without restarting from scratch.



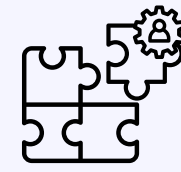
RESPOND - REPLAN - EXECUTE

Existing Solutions & Gaps



Existing Solutions

- ✦ Interrupted agents discard the entire in-memory graph and issue a fresh prompt to the LLM with updated chat history.
- ✦ When a new prompt arrives, the system attempts a blunt `task.cancel()` or `thread abort`.
- ✦ Agent operates in discrete Thought → Action → Observation turns.
Once tool invocation starts, the main reasoning thread blocks until network I/O completes.



Gaps

- ✦ User Lockout: Users are blocked for 3–5s while tools run.
No Mid-Turn Reactivity: Interruptions are queued rather than handled live
- ✦ No Stale Gates: Dispatched API calls finish in the background → causes phantom bookings & dirty writes.
No Risk Separation: Treats safe reads (search) the same as financial commits (pay).
- ✦ High Latency: 3–6s cold restart penalty for minor word changes.
Context Amnesia: Wipes valid intermediate results (cached hotels, routes) that could be reused.

Our Solutions & Architecture Diagram

1 Dual-Process Orchestration (Fast vs. Slow Path)

Fast Path (<1 ms): Instant WebSocket ACK + Intent Volatility Score (IVS) using a lightweight classifier.

Slow Path:Google Gemini builds a Plan DAG and launches speculative shadow branches to pre-fetch likely needs.

Result: Instant feedback without waiting for LLM/tools.

2 Version-Gated DAG Engine & Stale Gate

Each interruption increments the version V1 → V2.

Workers check is_stale() before execution and resolution; outdated results are discarded.

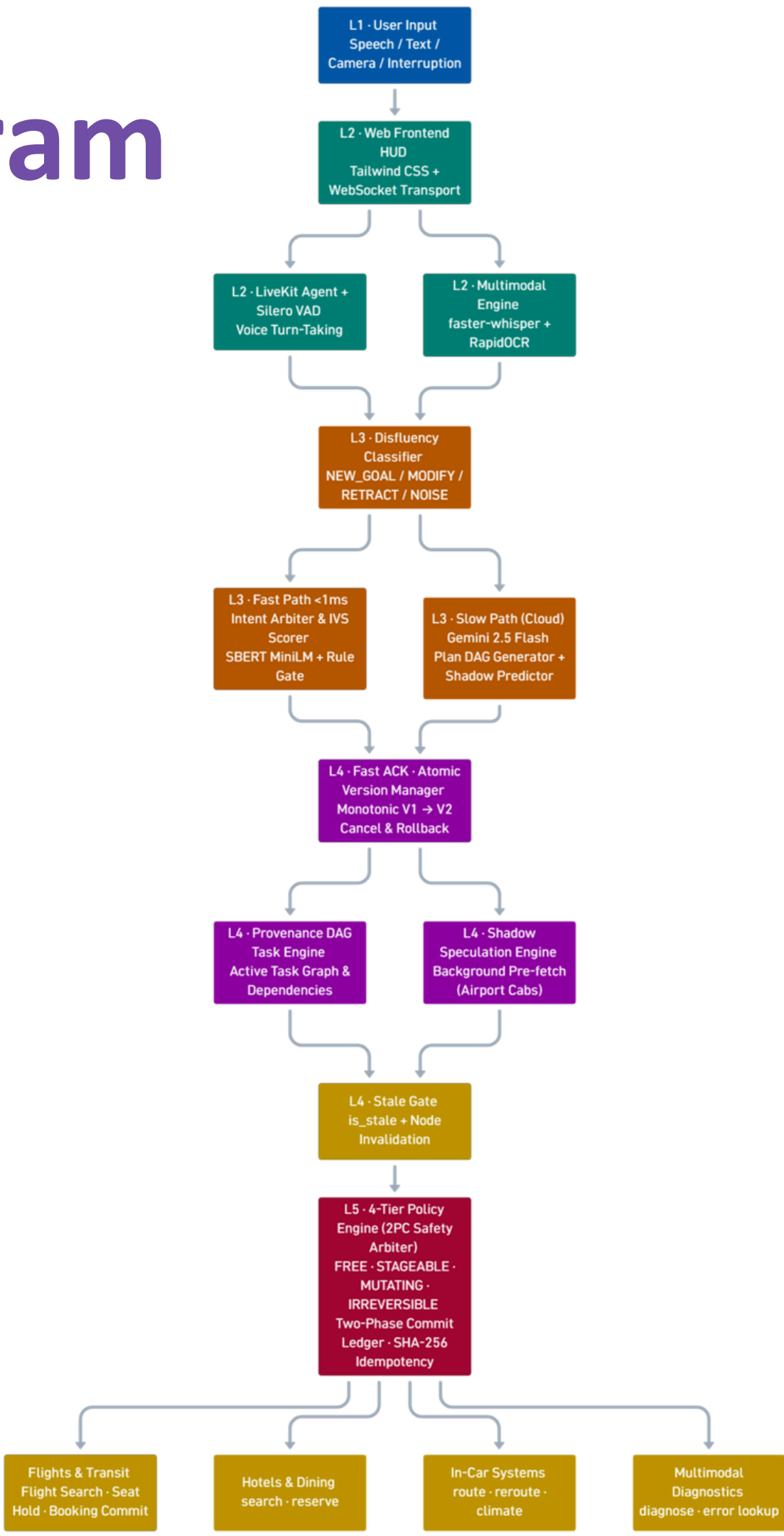
Surgical invalidation (<2 ms): Only affected downstream nodes are dropped; independent work is preserved.

3 Policy Matrix & 2-Phase Effect Ledger

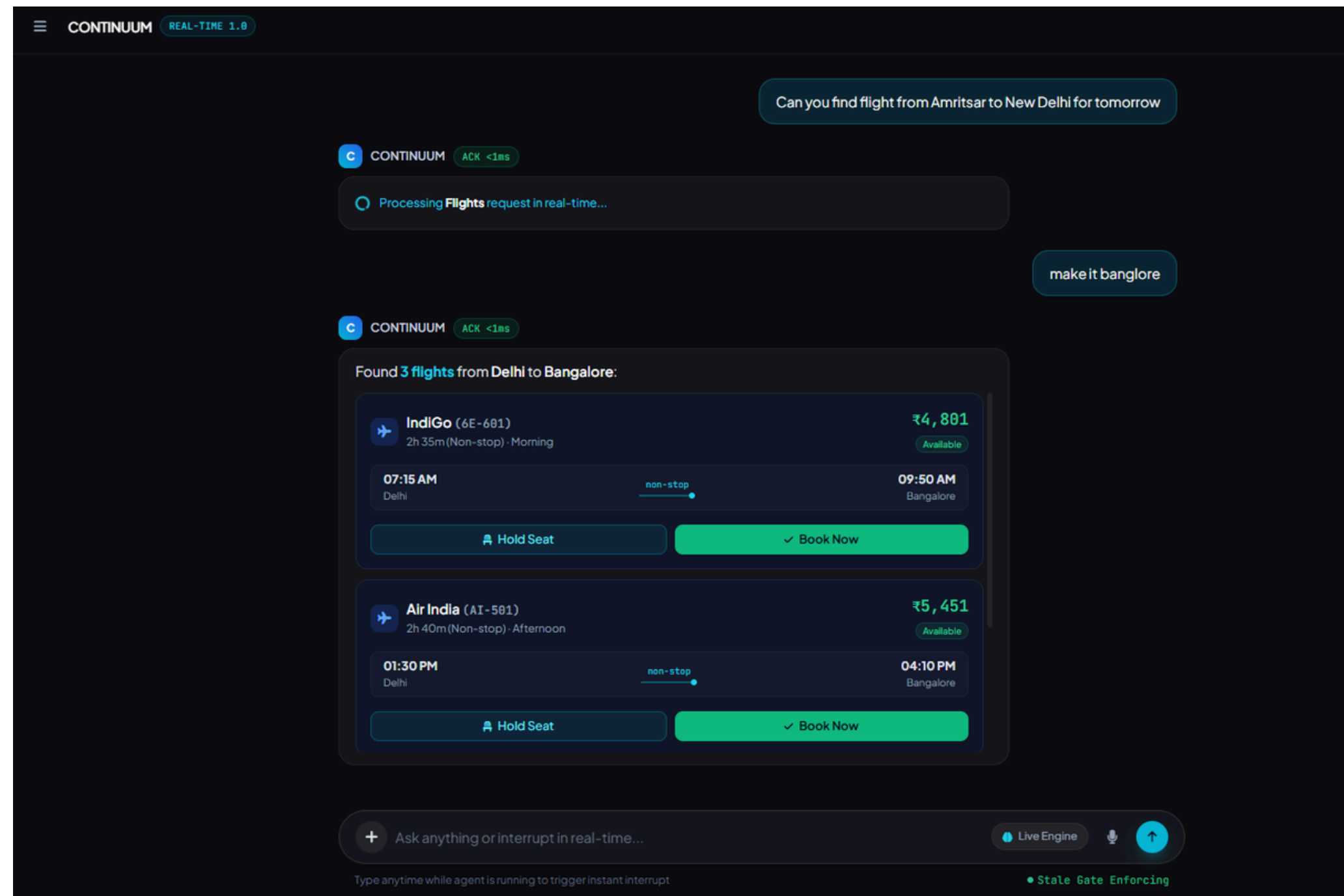
FREE: Read-only queries → Auto-execute.

STAGEABLE: Soft holds/drafts → Auto-stage.

MUTATING / IRREVERSIBLE: State changes & payments → User confirmation + low IVS (<0.60).



Demo & Product Walkthrough



🟡 **Mid-task interrupt** —
User switches Delhi → Bangalore while search is running

⚡ **ACK <1ms** —
System perceives the change instantly, no re-type needed

🛡️ **Stale Gate fires** —
Old Amritsar→Delhi result is silently dropped, never shown

✅ **Correct result served** —
Delhi→Bangalore flights returned with live pricing & booking tiers



Real interruption . Real correction . Zero confusion.

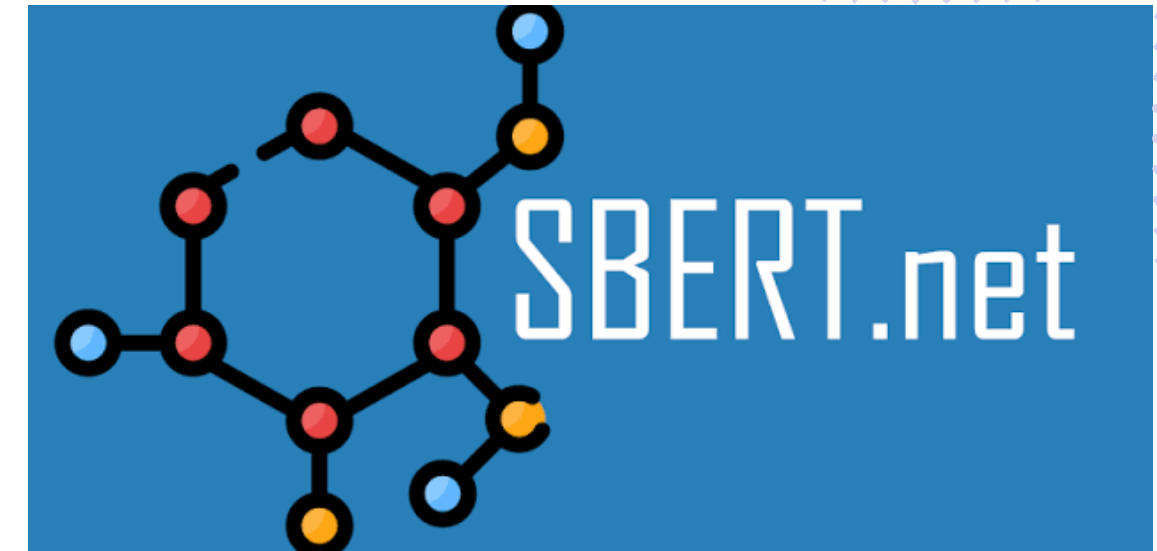
Tools and tech stack used

◆ **Backend & Concurrency:** Python 3.12, FastAPI, asyncio, WebSockets, LiveKit (WebRTC)

◆ **AI & NLP Stack:**
Google Gemini Flash (Open-Domain Reasoning), SentenceTransformers (MiniLM), PyTorch, 5-Way Intent Arbiter & IVS

◆ **Frontend:** Google Gemini Flash (Open-Domain Reasoning), SentenceTransformers (MiniLM), PyTorch, 5-Way Intent Arbiter & IVS

◆ **State & DAG Engine:** Version-Gated Provenance DAG Executor, Atomic Monotonic State Manager, Two-Phase SHA-256 Idempotency Ledger



Impact & Use case

SAMSUNG ECOSYSTEM DOMAINS



Galaxy AI & Bixby Voice

Full-duplex speech with barge-in
Users can correct speech mid-turn
Zero awkward conversational pauses



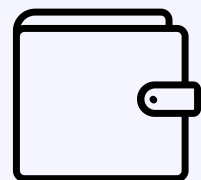
SmartThings & Home IoT

Coordinates multi-device routines
Prevents accidental appliance actions



Connected In-Car Infotainment

Hands-free driver navigation
Re-routes without losing climate tasks



Enterprise Commerce & Support

Resilient checkout workflows
Honest cancellation workflows



MEASURED USER & PLATFORM IMPACT



Up to 80% Faster Execution

Preserves valid background work instead of restarting.



Zero Duplicate Financial Actions

Effect Ledger prevents double-charging during timeouts



<1 ms Perceived Lag

Instant ACK confirms the user was heard.



50% Lower LLM API Overhead

Fused prompting reduces token and cloud overhead.

Innovation highlights, results and limitations

Innovation Highlights

1-Call Fused Delta Arbiter

Fuses diff extraction (Delta) and 5-way classification into a single prompt

4-Axis Typed Provenance:

Tracks freshness, capability, tool, verification

Selective Stale Gate

Drops obsolete background results from prior state versions (V1) without canceling still-valid work.

Manifest-Driven Generic Planner:

Zero-shot tool binding and dependency chaining for arbitrary JSON Schema manifests.

Technical Results

96.4% Held-Out Generalization

Achieved on 28 frozen held-out turns across unseen tool domains.

100.00 / 100 Runtime Score

Passed all 18 timed stress tests and 11 multimodal scenarios.

Ablation-Proven Gates

Removing the Stale Gate caused wrong-city results; removing timeout verification caused duplicate mutations.

308 Tests Passing

100% green CI across Python 3.10–3.12 with strict mypy typing.

Limitations

Cloud Credential Dependencies

LiveKit streaming verified through deterministic mocks; live rooms require cloud API keys.

Upstream Perception Grounding

Audio/video relies on ASR/OCR; confidence < 0.72 triggers clarification.

Session-Scoped Memory

V1→V2→V3 state is isolated per session; cross-session memory is planned for V2.

What's next

From prototype to a real-world, interrupt-aware assistant



01 · Data

Live API integration

Swap mock flights for Amadeus /
Skyscanner and real bookings



02 · Interaction

Voice interface

Full-duplex speech: interrupt and correct by
speaking



03 · Reach

Multi-domain expansion

Hotels, cabs, dining, and in-car assistants
on one engine



04 · Deployment

On-device deployment

Optimised for Samsung Galaxy edge
devices, works offline

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India

